

Cost-reduction options

All dollar figures are planning estimates at INR95.2236/USD, based on the attached model. They exclude staff unless explicitly stated. "Current" means the PDF's modeled production footprint, not an invoice.

Option matrix

Driver	Low-risk existing architecture	Moderate change	Aggressive redesign	Recommendation
RDS/proxy/replica	Prove replica unused; remove it; right-size Redis after Performance Insights. \$915 -> ~\$600 fixed; save ~\$315/mo; S-M effort, low migration risk.	Staging-class DB/cache, one task, conditional NAT/endpoints; \$915 -> ~\$204; save ~\$711; M effort, restore/SLO risk.	Single VM/low-cost Postgres; \$915 -> ~\$45; save ~\$870; L ops/backup/security burden and lower HA.	Moderate for beta; do not self-host for growth.
Gemini	Trim full listing JSON/tool schemas; cache stable context; \$45.70 -> ~\$23; save ~\$23; S-M, low product impact.	Route classification/search to cheaper model; \$45.70 -> \$10-\$18; M, quality regression risk.	Deterministic search/booking, AI fallback only; \$45.70 -> ~\$4.60; L, less differentiation.	Low-risk now, moderate after quality tests.
Razorpay	Negotiate rate after volume; \$151.80 -> ~\$130; S, low product impact.	Transparent fee/commission covers MDR; vendor cost remains but contribution turns positive; M, conversion/tax risk.	UPI-first/multi-PA routing; \$151.80 -> \$60-\$100; L, reconciliation/compliance risk.	Pricing fix is mandatory before charging.
Firebase OTP	Resend/device/user quotas and App Check; 10K modeled OTP \$27 -> ~\$19; S, low impact.	Email/social/passkey for repeat auth; \$27 -> ~\$6; M, recovery redesign.	Phone only at risk/KYC checkpoint; \$27 -> ~\$2; M-L, conversion tradeoff.	Quotas now; A/B test alternatives.
Google Places	Debounce/session tokens/cache/minimum chars; high-scale \$840 -> ~\$588; S, low impact.	India city/place dataset with Google only for precise address; \$840 -> ~\$168; M, data maintenance.	Self-host geocoder; \$840 -> \$100-\$400 infra; L, freshness/licence/ops.	Moderate only after volume.
Media/moderation	Size limits, compression, hashes and quotas; 20-30% variable saving; S.	Async queue/dedupe and publish-after-review; 50-70% peak saving; M, delayed listing.	Self-host consolidated workers; scale-dependent; L model-update/security burden.	Low-risk immediately; async at meaningful volume.
Logs/events	Redact/noisy debug, sampling and TTLs while retaining CERT-In 180-day protected logs; 20-60% ingestion saving; S.	Separate compliance archive from hot analytics and singleton rollups; M.	Open-source observability stack; service cost falls but ops rises; L.	Retention must not be reduced below legal requirement.

Current vs target architecture

Architecture	Fixed/variable profile	Security impact	Operational burden	Lock-in	Break-even
Current production	\$915 fixed + ~\$202 variable at 1K	Good intent, but public origin/Redis gaps	High	AWS/vendor high	Not viable at INR3 fee
Optimised current	~\$204 fixed + ~\$188 variable after AI/payment controls	Preserve private DB/backups; test restore	Medium	AWS high	~500-2,000 MAU depending commission
Alternative managed	~\$100 fixed + same variable using managed Postgres/serverless cache/CDN	Vendor controls/DPA must be verified	Medium	Moderate/high	Lower than current; migration ~\$6k INR-equivalent planning
Open-source lower-cost	~\$45 fixed + same variable	More patch/backup/incident risk	Very high; 0.5+ FTE realistic	Lower vendor, higher people dependency	Only if traffic justifies ops team

TCO warning

At 1K MAU, a self-hosted \$45/month invoice is not a \$45/month product. Add Indian engineering/on-call, backups, security testing, incident response and support: a realistic 12-month TCO can exceed the optimized managed option. Keep the managed path until steady GMV and traffic justify a dedicated platform team.

Recommended target

Target $\leq$ INR25,000 total direct vendor cost per 1,000 MAU/month including modeled gateway fees, and $\leq$ INR10,000 excluding pass-through MDR, with a hard budget alert at 120% and an automated kill switch for AI/Maps/voice. This is commercially sensible because it leaves room for support, tax and fraud while India-facing booking commissions are tested.

Findings

ID	Domain	Title	Severity	Confidence	Evidence	User/business impact	Recommended correction	Effort	Owner	Release blocker	Status
COST-001	Architecture	Remove idle fixed capacity before growth	High	High confidence	r6g replica/Redis/proxy/NAT/endpoint topology	Cash burn	Staged right-sizing experiment with restore test	Medium	CTO/FinOps	yes commercially	Open
COST-002	AI	Prompt/tool context is the controllable variable cost	High	Confirmed	PDF 97% input-token spend	Bill cliff	Context trimming/model routing	Medium	AI lead	before 1K	Open
COST-003	Abuse	Expensive unauthenticated proxies need hard budgets	High	Confirmed/high confidence	Maps/translation/AI/voice routes and limiter order	Denial-of-wallet	Verified subject quotas, budgets and circuit breakers	Medium	SRE	yes	Open
COST-004	Ops	Open-source self-hosting hides people cost	Medium	High confidence	TCO model	False savings	Compare full TCO, not invoice	Low	CFO	no	Open